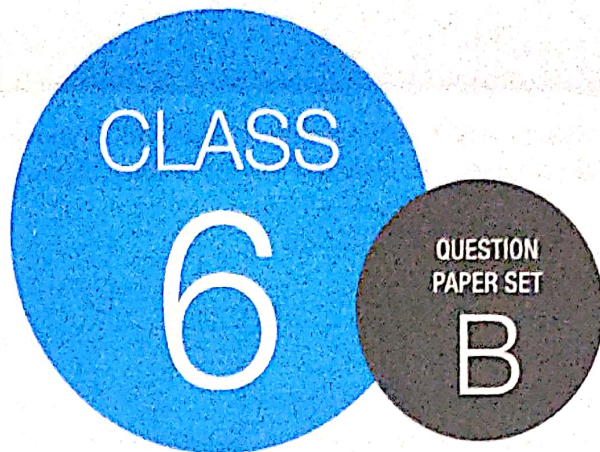




**SOF INTERNATIONAL
MATHEMATICS OLYMPIAD
2017-18**



DO NOT OPEN THIS BOOKLET UNTIL ASKED TO DO SO

Total Questions: 50 | Time: 1 hr.

Name:.....

Section:..... SOF Olympiad Roll No.:..... Contact No.:.....

Guidelines for the Candidate

1. You will get additional ten minutes to fill up information about yourself on the OMR Sheet, before the start of the exam.
2. Write your **Name, School Code, Class, Section, Roll No.** and **Mobile Number** clearly on the **OMR Sheet** and do not forget to sign it. We will share with you your marks / result on your mobile number.
3. The Question Paper comprises four sections:

Logical Reasoning (15 Questions), **Mathematical Reasoning** (20 Questions), **Everyday Mathematics** (10 Questions) and **Achievers Section** (5 Questions)

Each question in Achievers Section carries 3 marks, whereas all other questions carry one mark each.

4. All questions are compulsory. There is no negative marking. Use of calculator is not permitted.
5. There is only ONE correct answer. Choose only ONE option for an answer.
6. To mark your choice of answers by darkening the circles on the OMR Sheet, use **HB Pencil** or **Blue / Black ball point pen** only. E.g.
Q.16: Rahul bought 4 kg 90 g of apples, 2 kg 60 g of grapes and 5 kg 300 g of mangoes. The total weight of all the fruits he bought is _____.
A. 11.450 kg B. 11.000 kg C. 11.350 kg D. 11.250 kg

As the correct answer is option A, you must darken the circle corresponding to option A in the OMR Sheet.

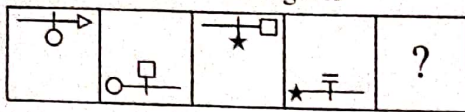
16. ● B C D

7. Rough work should be done in the blank space provided in the booklet.
8. Return the OMR Sheet to the invigilator at the end of the exam.
9. Please fill in your personal details in the space provided on this page before attempting the paper.

LOGICAL REASONING

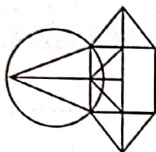
1. Select a figure from the options which will continue the same series as established by the Problem Figures.

Problem Figures



- A. B. C. D.

2. Count the number of triangles in the given figure.



- A. 10
B. 12
C. 14
D. 16

3. If the digits of the following numbers are reversed and then the numbers are arranged in ascending order, then what will be the middle digit of the middle number?

254 439 671 894 958

- A. 9
B. 5
C. 3
D. 7

4. Find the missing number, if same rule is followed in all the three figures.

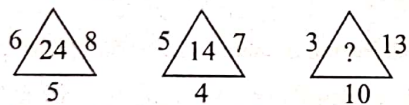


Figure (1) Figure (2) Figure (3)

- A. 39
B. 26
C. 35
D. 25

5. Select the CORRECT water image of the given combination of letters.

COMBINATION

- A. COWBIVLION
B. CWOVIVLION
C. COBIVLION
D. COWBIVLION

6. Select a figure from the options in which Fig. (X) is exactly embedded as one of its parts.



Fig. (X)

- A. B. C. D.

7. If 'Clock' is called 'Television', 'Television' is called 'Radio', 'Radio' is called 'Oven', 'Oven' is called 'Grinder' and 'Grinder' is called 'Iron', then in which will a woman bake?

- A. Radio
B. Oven
C. Grinder
D. Iron

8. Which of the following options would complete the pattern given in Fig. (X)?

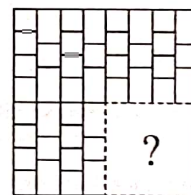
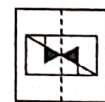


Fig. (X)

- A. B. C. D.

9. A square transparent Sheet (X), with a pattern and a dotted line on it is given. Find the figure from the options as to how the pattern would appear when the transparent sheet is folded along the dotted line.



Sheet (X)

- A. B. C. D.

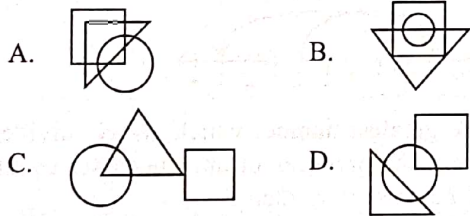
10. Vansh and Puneet started walking from two different positions A and B respectively. Vansh walks 3 km North and turns to East and walks 4 km and again turns to North and walks 2 km and finally turns to East and walks 4 km to reach point 'C'. Similarly, Puneet walks 3 km North and turns to West and walks 4 km and finally turns to North, walks 2 km and meet Vansh at point 'C'. What was the distance between Vansh and Puneet's starting positions?

A. 10 km
B. 12 km
C. 15 km
D. 14 km

11. If 'P' denotes ' $\times$ ', 'R' denotes '+', 'M' denotes '-' and 'S' denotes ' $\div$ ', then the value of $24R8S7M2P5$ is

A. $4\frac{4}{5}$
B. 28
C. 40
D. 0

12. Select a figure from the options which satisfies the same condition of placement of the dots as in Fig. (X).



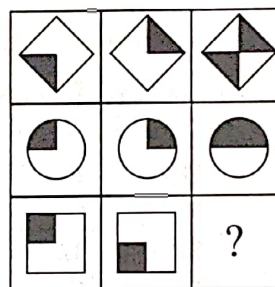
13. Arrange the given words as they occur in a dictionary and select the correct option.

1. Fight 2. Freak
3. Faint 4. Fault
5. Flick
A. 4, 3, 1, 5, 2
B. 4, 3, 2, 5, 1
C. 3, 4, 1, 5, 2
D. 3, 4, 5, 1, 2

14. Pointing to a girl in the photograph, Amar said, "Her mother's brother is the only son of my mother's father". How is the girl's mother related to Amar?

A. Mother
B. Sister
C. Aunt
D. Grandmother

15. Select a figure from the options which will complete the given figure matrix.



MATHEMATICAL REASONING

16. Find the estimated difference between 16928 and 8952 by rounding off each number to nearest hundreds.

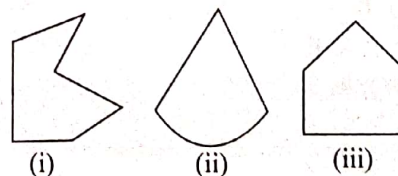
A. 7900
B. 8000
C. 7800
D. 7976

17. Which of the following statements is CORRECT?

A. The product of two negative integers is always less than both the integers.
B. The additive inverse of a negative integer is negative.
C. Multiplicative inverse of 5 is $\frac{1}{5}$.

- D. The difference between an integer and its additive inverse is always odd.

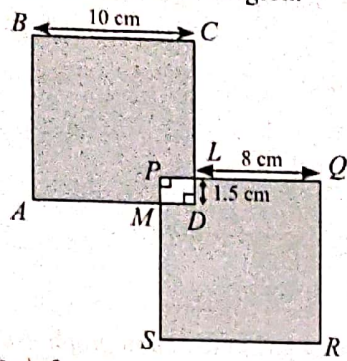
18. Study the given figures carefully.



Which of the above figures are polygons?

A. Only (i)
B. Both (i) and (ii)
C. Both (i) and (iii)
D. (i), (ii) and (iii)

19. If $ABCD$ and $PQRS$ are two identical squares, then find the area of the shaded region.



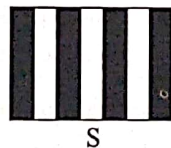
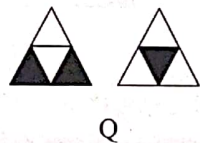
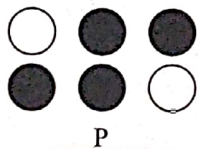
- A. 180 cm^2
 B. 170 cm^2
 C. 194 cm^2
 D. 173 cm^2
20. Find the measure of smaller angle formed by the hour and the minute hands of a clock at 10 O' clock.

- A. 60°
 B. 90°
 C. 45°
 D. 30°

21. A triangular prism has P faces, Q vertices and R edges.

	P	Q	R
A.	5	5	8
B.	6	8	12
C.	5	6	8
D.	5	6	9

22. Arrange the shaded fraction of following figures in ascending order.

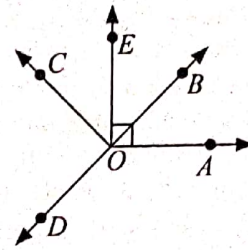


- A. Q, P, S, R
 B. P, Q, R, S
 C. S, R, Q, P
 D. R, S, P, Q

23. Select the CORRECT match.

- A. Twelve and thirty nine thousandths – 12.420
 B. Four and forty hundredths – 4.004
 C. Sixteen and two tenths – 16.2
 D. Eight and five hundredths – 80.50

24. Study the given figure carefully. If DOB is a straight line, then match the columns.



Column-I

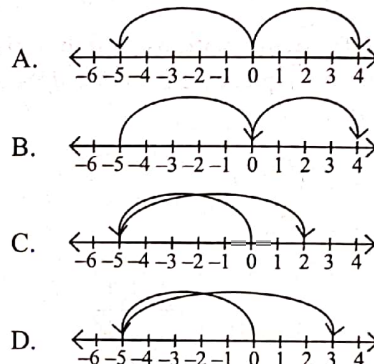
- (a) $\angle AOB$
 (b) $\angle AOE$
 (c) $\angle AOC$
 (d) $\angle BOD$

Column-II

- (i) Straight angle
 (ii) Acute angle
 (iii) Right angle
 (iv) Obtuse angle

- A. (a)→(i); (b)→(iv); (c)→(ii); (d)→(iii)
 B. (a)→(ii); (b)→(i); (c)→(iii); (d)→(iv)
 C. (a)→(ii); (b)→(iii); (c)→(iv); (d)→(i)
 D. (a)→(i); (b)→(ii); (c)→(iii); (d)→(iv)

25. Which of the following number lines correctly represents $(-5) + 8$?



26. Find the greatest number which always divides the product of the predecessor and successor of an odd natural number other than 1.

- A. 6
 B. 4
 C. 8
 D. 16

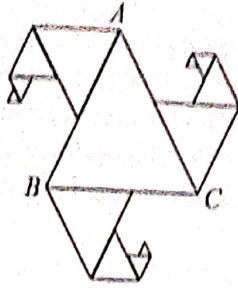
27. Amulya and Amar visited two places A and B respectively in Kashmir and recorded the minimum temperatures on a particular day as -4°C at A and -1°C at B. Which of the following statements is true?

- A. A is cooler than B.
 B. B is cooler than A.
 C. There is a difference of 2°C in the temperature.
 D. The temperature at A is 4°C higher than that at B.

28. The number of diagonals in a heptagon is _____.

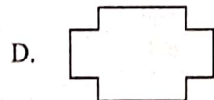
- A. 14
 B. 15
 C. 18
 D. 10

29. In the given figure, all triangles are equilateral and $AB = 16$ units. Other triangles have been formed by taking the mid points of the sides. What is the perimeter of the figure?



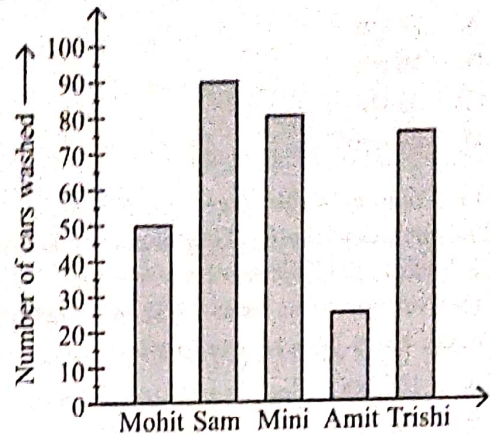
- A. 92 units
B. 93 units
C. 91 units
D. None of these
30. Find the value of $LVIII + XXIV + LXXXIX + XXXII - XCIV$.
- A. XCIX
B. XCVIII
C. CIX
D. CVII

31. Which of the following figures has more than three lines of symmetry?



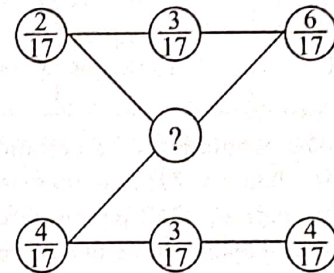
32. The number 7250371 is written in International System as _____.
- A. Seventy two lakh fifty thousand three hundred seventy one.
B. Seven million twenty five thousand three hundred seventy one.
C. Seven million two hundred fifty thousand three hundred seventy one.
D. None of these

33. The given bar graph shows number of cars washed by five children to raise money for local children hospital.



What is the difference between the number of cars washed by (Trishi and Sam) & (Mohit and Mini)?

- A. 20
B. 30
C. 35
D. 25
34. What should come in the place of '?' so that the sum of fractions on each line is same?



- A. $\frac{1}{17}$
B. $\frac{3}{17}$
C. $\frac{5}{17}$
D. $\frac{6}{17}$
35. The sum of predecessor and successor of the smallest five digit number formed from the digits 1, 4, 0, 2, 5 (each digit used only once) is _____.
- A. 20490
B. 20480
C. 2490
D. 20508

EVERYDAY MATHEMATICS

36. Team coach Abhimanyu needs to buy 16 bats for the cricket team. The list below shows the prices of cricket bats at two shops.
- (i) Shop A : A package of 8 cricket bats costs ₹ 2560.
(ii) Shop B : A package of 4 cricket bats costs ₹ 1550.

What is the amount of money he will save if he buys 16 cricket bats from shop A than shop B?

- A. ₹ 2000
B. ₹ 1080
C. ₹ 1200
D. ₹ 1000

37. A rectangular piece of land measures 4.5 cm by 2.5 cm. Each side is to be fenced with four rows of wires. What is the total length of wire needed?

A. 28 cm
B. 56 cm
C. 35 cm
D. 42 cm

38. In a regular week, there are 5 working days and for each day, the working hours are 8. A man gets ₹ 2.40 per hour for regular work and ₹ 3.20 per hour for overtime. If he earns ₹ 432 in 4 weeks, then how many hours does he work for?

A. 160 hrs
B. 175 hrs
C. 180 hrs
D. 195 hrs

39. To make a mocktail, Ashima mixed $2\frac{1}{3}$ L of soda, $1\frac{2}{3}$ L of lime syrup and $1\frac{5}{6}$ L of water together. Find the total amount of mocktail she made.

A. $5\frac{5}{6}$ L B. $2\frac{1}{6}$ L
C. $4\frac{1}{6}$ L D. $5\frac{1}{3}$ L

40. In 2016, the population of a certain village was 105250. By January 2017, it increased by 4315. During this period, 9242 people left the village. What was the population of the village in January 2017?

A. 100323
B. 118807
C. 110177
D. None of these

41. Three boys steps off together from the same spot. Their steps measure 63 cm, 70 cm and 77 cm, respectively. What is the minimum distance each should cover so that all can cover distance in complete steps?

A. 6930 cm
B. 6000 cm
C. 7000 cm
D. 6520 cm

42. It was estimated that because of people switching to Metro trains, about 33000 tonnes of CNG, 3300 tonnes of diesel and 21000 tonnes of petrol was saved by the end of year 2016. Find the fraction of the quantity of diesel and petrol is saved together to the quantity of CNG saved.

A. $\frac{1}{10}$ B. $\frac{81}{110}$
C. $\frac{7}{11}$ D. $\frac{110}{81}$

43. Raghav walked 42.25 km in four days. He walked 8.25 km on Monday, 7.52 km on Tuesday, 11.27 km on Wednesday and rest of distance on Thursday. How much distance did he walk on Thursday?

A. 16.24 km
B. 16.41 km
C. 15.21 km
D. 15.41 km

44. A barrel has 70 L 200 mL of milk. Jugs, each of capacity 130 mL are filled from this barrel. How many such jugs can be filled from this barrel?

A. 500
B. 540
C. 480
D. 620

45. A vendor supplies 105 litres of milk to a hotel in the morning and 85 litres of milk in the evening. If the milk costs ₹ 20 per litre, then how much money is due to the vendor per week?

A. ₹ 35000
B. ₹ 42000
C. ₹ 39500
D. ₹ 26600

ACHIEVERS SECTION

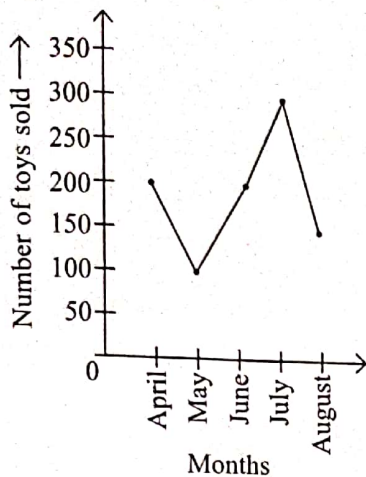
46. Fill in the blanks.

- (i) Two lines having no common point are called P lines.
(ii) The minimum number of lines required to form a closed figure is Q.
(iii) A chord that passes through the centre of a circle is R.

- (iv) $\frac{3}{5}$ of a right angle is S.

	P	Q	R	S
A.	Intersecting	2	diameter	90°
B.	Parallel	3	radius	70°
C.	Intersecting	4	segment	60°
D.	Parallel	3	diameter	54°

47. The given line graph shows the number of toys sold by a store in five months.



What is the ratio of the toys sold in April and June together to the toys sold in August and July together?

- A. 9 : 8
B. 8 : 9
C. 7 : 6
D. 6 : 7

48. Read the statements carefully and state 'T' for true and 'F' for false.

- (P) A fraction is said to be in simplest form if its numerator and denominator have no common factor other than 1.
(Q) $\frac{5}{9}$ is greater than $\frac{4}{5}$.
(R) Fractions can't be represented on the number line.
(S) In the place value of decimals, as we go from left to right, the multiplying factor becomes $\frac{1}{10}$ of the previous factor.

(P) (Q) (R) (S)

- A. T F F T
B. T T F T
C. F T F F
D. F T T F

49. Study the given statements carefully and select the correct option.

Statement-I : A natural number is divisible by 8, if the number formed by last three digits is divisible by 8.

Statement-II : 987648 is divisible by 8.

- A. Both Statement-I and Statement-II are true.
B. Both Statement-I and Statement-II are false.
C. Statement-I is true but Statement-II is false.
D. Statement-I is false but Statement-II is true.

50. Match the following.

Column-I

Column-II

- (P) The smallest integer greater than every negative integer is (i) 6
(Q) The value of $49 - (-40) - (-3) + 69 - 80$ is (ii) 56
(R) If the number 7254 * 98 is divisible by 22, the digit at * is (iii) 0
(S) A wire is cut into several small pieces. Each of the small pieces is bent into a square of side 2 cm. If the total area of small squares is 28 cm^2 , then the original length of wire (in cm) is (iv) 81
- A. (P) - (i), (Q) - (ii), (R) - (iii), (S) - (iv)
B. (P) - (iii), (Q) - (ii), (R) - (i), (S) - (iv)
C. (P) - (iii), (Q) - (iv), (R) - (i), (S) - (ii)
D. (P) - (iii), (Q) - (iv), (R) - (ii), (S) - (i)

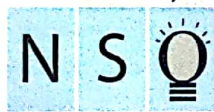
SPACE FOR ROUGH WORK



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